Kamal Karki

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PROFILE

Aerospace engineer at **Airbus** with research interests in **robotics, robot dynamics, motion control, and mechatronic systems**. Contribute to the open-source **Robotics Toolbox for Python**, with work on manipulator inverse kinematics and rigid-body dynamics, complemented by hands-on robot development spanning mechanism design, electronics integration, and **PID-based speed control** through ABU Robocon. Seeking thesis-based graduate research opportunities in Design, robotics and control within Mechanical/Aerospace Engineering and related engineering departments.

EDUCATION

Pulchowk Engineering Campus, Tribhuvan University **May 2021 – Apr 2025**
Bachelor of Engineering in Aerospace Engineering *Lalitpur, Nepal*

- **Overall: 74.16% (First Division), 3.75/4 (Scholaro Estimate)** | **Final Semester: 83.8%** | **All Nepal Engineering Entrance Rank: 419/18,000+ (Top 2.5%)**
- **Relevant Coursework:** Control Systems, Engineering Dynamics, Theory of Machines, Numerical Methods, Finite Element Method, Continuum Mechanics, Instrumentation & Sensors, Fault Monitoring & Diagnosis, Unmanned Air Vehicle Synthesis, Computer Aided Design & Manufacturing

Kantipur Higher Secondary School **2016 – 2018**
Higher Secondary Education (+2), Science *Biratnagar, Nepal*

- **GPA: 3.78 / 4.00** | Awarded a **National Examinations Board (NEB) scholarship**

SCHOLARSHIPS & AWARDS

- **IOE TU Merit Full Scholarship**, Tribhuvan University – Awarded for B.E. in Aerospace Engineering at Pulchowk Engineering Campus.
- **NEB Scholarship for Academic Excellence**, National Examinations Board, Nepal – Awarded for higher secondary education (+2 Science).
- **Nagase Award for Best Engineering Performance**, ABU Robocon 2022, New Delhi – Team placed **3rd overall** in the Asia-Pacific region.

OPEN-SOURCE CONTRIBUTIONS

Robotics Toolbox for Python | *Open-Source Contributor* **2026 – Present**

- Contributor to the open-source companion library to Peter Corke's *Robotics, Vision and Control*, covering manipulator kinematics, dynamics and motion planning; pull requests reviewed and merged by the author.
- Fixed **rigid-body dynamics** for URDF robot models with rotated inertial frames by transforming inertia tensors into the link frame ($I_{\text{link}} = R\, I\, R^{\top}$), correcting inverse dynamics, mass-matrix and Coriolis computations.
- Corrected the **Gauss–Newton inverse-kinematics solver** to solve the weighted normal equations $(J^{\top}WJ)^{-1}J^{\top}We$, restoring task-space error weighting.
- Redesigned **URDF/xacro robot-model loading** to report the failing stage, XML element and line through a structured exception, backed by 13 regression tests.

PyBaMM – Python Battery Mathematical Modelling | *Open-Source Contributor* **2026 – Present**

- Diagnosed defects in the **battery degradation models** (SEI growth, lithium plating), including a numerical failure affecting several published parameter sets; submitted fixes with regression tests.
- Worked with model authors and core maintainers on the fix design; the findings led to a package-wide redesign of model-option validation.

SELECTED RESEARCH & ENGINEERING PROJECTS

Investigation of the effects of Printing Parameters and Reinforcement on the Mechanical Properties of 3D-Printed PLA+ Composites | *FDM, Materials Testing, Machine Learning* **Jun 2024 – Feb 2025**

- Tested how nozzle temperature, infill pattern and infill density affect 3D-printed PLA+ (**ASTM D638** tensile, compression and damping tests); rule-of-mixtures analysis indicated **negligible load transfer** from embedded steel-wire reinforcement ($\eta \approx 0$ for two- and three-wire configurations).
- Built an **ML pipeline** pooling **9 open FDM datasets (172 specimens)** and benchmarked **20 regressors** under nested cross-validation; an **ARD Gaussian process** predicted tensile strength to **RMSE 6.2 MPa (R^2 0.90)**.
- Model interpretation (SHAP, partial dependence) identified **infill density** as the dominant process effect; cross-laboratory validation showed published datasets captured **strength trends** but required calibration specimens for absolute predictions.

Robotics Club, Pulchowk Campus **Mar 2022 – Jan 2023**
Robotics Engineer *Lalitpur, Nepal*

- Contributed across the **end-to-end development of Robot A (Thrower)** for ABU Robocon 2022, spanning mechanical design, fabrication, electronics, PCB development, and integration of the robot's hardware and control systems.
- Designed and built a **dual-rotational throwing mechanism** and implemented **PID-based speed control** to regulate its rotational speed, combining mechanical mechanism design with feedback control.
- Developed and integrated electronic hardware and contributed to the robot's **leg-control and motion systems**, using Raspberry Pi hardware as part of the robot's control setup.
- Contributed to the team that secured **third place in the Asia-Pacific region** and received the **Nagase Award for Best Engineering Performance** at ABU Robocon 2022 in New Delhi, India.

SWIFT: Conceptual Design of a Mid-Range Commercial Aircraft | *Departmental Poster Presentation* **Mar 2024 – Apr 2024**

- Co-designed a **30-passenger, twin-turbofan aircraft** with a **5,000 km design range**, performing preliminary weight estimation, wing and tail sizing, airfoil selection, and landing-gear configuration.
- Evaluated **flight performance and dynamic stability** using analytical calculations and simulation, analyzing short-period, phugoid, Dutch-roll, and spiral modes and assessing handling qualities with a **Cooper–Harper rating of 3**.

Effect of Sweep Angle on Missile Canard Aerodynamics | *CATIA V5, ANSYS CFD* **Oct 2023 – Dec 2023**

- Conducted **CFD analysis in ANSYS** of canard fins modelled in CATIA V5 across sweep angles of **40°–80°** to characterize lift and drag behaviour.
- Identified an **optimal sweep-angle range** that maximizes lift without a disproportionate drag penalty.

Blended-Wing-Body UAV: Design, Fabrication, and Flight Test | *CATIA V5* **Sep 2021 – Nov 2021**

- Designed and modeled a blended-wing-body UAV airframe using **CATIA V5**, integrating power systems, motors, and an engine for optimal aerodynamic stability.
- Configured custom **Arduino**-based flight controllers and avionics, conducting extensive ground and flight testing to validate control response and system performance.

ENGINEERING EXPERIENCE

Airbus India Private Limited **Feb 2026 – Present**
Associate Engineer – Design *India*

- Design, develop and test **aircraft cabin** using **CATIA**, performing tolerance checks, 3D assembly interference analysis, and clearance assessments to resolve component-fit issues.
- Lead a **sub-team of two engineers** in feasibility studies and root-cause analysis of manufacturing issues, coordinating design modifications with manufacturing teams and assessing manufacturability.
- Work across the aircraft component lifecycle using **VPM and ENOVIA PLM tools, including PASS and TREND**, supporting configuration management, engineering-change workflows, assembly checks, and weight-related analyses.
- Developed an **ML-based aircraft design-reuse system** using **DBSCAN clustering** and priority-based ranking to retrieve prior designs for aircraft customization, with estimated savings of **3,500+ engineering hours annually**.

Airbus India Private Limited **Aug 2025 – Jan 2026**
Engineering Intern – Flight Warning Systems *India*

- Contributed to the **development and validation of A320/A321 Flight Warning System logic**, evaluating warning activation, suppression, clearing, sequencing, and priority across operating conditions and combinations of aircraft-system inputs.
- Prepared validation scenarios and analyzed system responses against expected behavior, investigating discrepancies across multiple **Flight Warning System functions** and their interfaces with other aircraft systems.
- Investigated **Flight Warning System defects** by reproducing anomalous warning behavior and examining associated input conditions and system logic to support failure analysis and technical investigation.
- Contributed to the **failure analysis of a critical aircraft electrical-system issue in the lavatory area**, examining associated Flight Warning System indications and warning logic to support the technical investigation.
- Developed a **YOLOv5-based computer-vision tool** to convert legacy **SAO logic-circuit diagrams** used in aircraft fault analysis into interactive web-based circuits, improving defect-analysis efficiency by **13%**.

Cloud Pro AI **Feb 2025 – Jul 2025**
Backend Engineer *Remote*

- Designed and deployed the **backend system from scratch** using **FastAPI and PostgreSQL**, implementing database schemas, APIs, session management, and secure JWT/OAuth2 authentication and authorization.
- Improved API responsiveness through **database indexing and asynchronous SQLAlchemy**, managing database schema evolution with **Alembic migrations**.

Buddha Air Pvt. Ltd. **Dec 2024**
Aircraft Maintenance Intern *Kathmandu, Nepal*

- Analyzed **ATR-42/72 flight and engine-monitoring data** across hundreds of recorded parameters to identify abnormal trends and system deviations supporting **preventive maintenance**.
- Performed **technical inspection, fault isolation, and corrective maintenance** on aircraft systems, including wheel/brake assemblies and pneumatic de-icing systems.

ORION Space **May 2023 – Jun 2023**
Research Engineering Intern *Bhaktapur, Nepal*

- Designed mechanical components for a modular **PocketQube Training Kit**, translating satellite-system concepts into a hands-on platform for space-engineering education in Nepal.
- Developed and iterated **designs**, fabricated prototypes through **3D printing**, and performed fit, assembly, and functional testing to refine geometry and manufacturability.

ENTREPRENEURSHIP & TECHNOLOGY INITIATIVES

Loksewa Sandhi **2024**
Co-Founder / Product Developer *Nepal*

- Developed and launched an **EdTech mobile app focused on government service examinations** using **Flutter** and **Node.js**.
- Delivered a **low-latency video service**, growing the platform to **3,000+ users**.

Beemalaya **2024**
Co-Founder / Product Developer *Nepal*

- Co-founded Beemalaya, an **insurance aggregator platform for Nepal** that lets users compare and access insurance products in one place.
- Raised **NPR 1.1M in angel funding** to build and scale the platform.
- Led technical execution, implementation, and user-oriented product iterations.

LEADERSHIP

Maths and Physics Society (MAPS), Pulchowk Campus **Jul 2022 – Dec 2024**
Vice President (Dec 2023 – Dec 2024) | Treasurer (Jul 2022 – Dec 2023) *Lalitpur, Nepal*

- Led strategic planning for society initiatives and organized the **Math and Physics Olympiad and Integration Bee**, attracting **400+ participants from across Nepal**.
- Raised **NPR 900K in sponsorship** from startups, companies, and organizations to fund the Olympiad.

Saakshar Nepal **Mar 2023 – Mar 2024**
Founding President *Lalitpur, Nepal*

- Founded Saakshar Nepal, a social organization delivering robotics and engineering education to **2,000+ students across 15+ schools in six districts**.
- Developed and led **hands-on robotics workshops** introducing fundamental engineering concepts, improving STEM accessibility for students below Class 8.

TECHNICAL SKILLS

Robotics & Control: Robot Kinematics and Dynamics, Mechanism Design, PID-Based Speed Control, URDF/xacro Parsing

Mechanical Design & PLM: CATIA V5, Solidworks, VPM, ENOVIA PLM (PASS, TREND), Assembly Interference and Clearance Analysis, Tolerance Checks, Configuration Management

Embedded Systems & Hardware: Raspberry Pi, PCB Development, Electronics Integration, Prototype Fabrication

Programming & Scientific Computing: Python, MATLAB, Git/GitHub, pytest, FastAPI, PostgreSQL, LaTeX

Machine Learning & Data Analysis: scikit-learn, DBSCAN Clustering, Gaussian Process Regression, Neural Networks, YOLOv5, SHAP, Nested Cross-Validation, pandas

Aerospace Systems & Engineering Analysis: Aircraft Systems, Avionics, Flight Warning Systems, System Verification and Validation, Failure Analysis, Aircraft Performance and Stability Analysis

SELECTED TRAINING

- **Neural Networks and Deep Learning** – DeepLearning.AI Mar 2024
- **Supervised Machine Learning: Regression and Classification** – DeepLearning.AI Jan 2024